COMPASS_TEMPEST_SGW_2024: November w/ original standard curve

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##Set Up

Read in first data file and assess standard curves

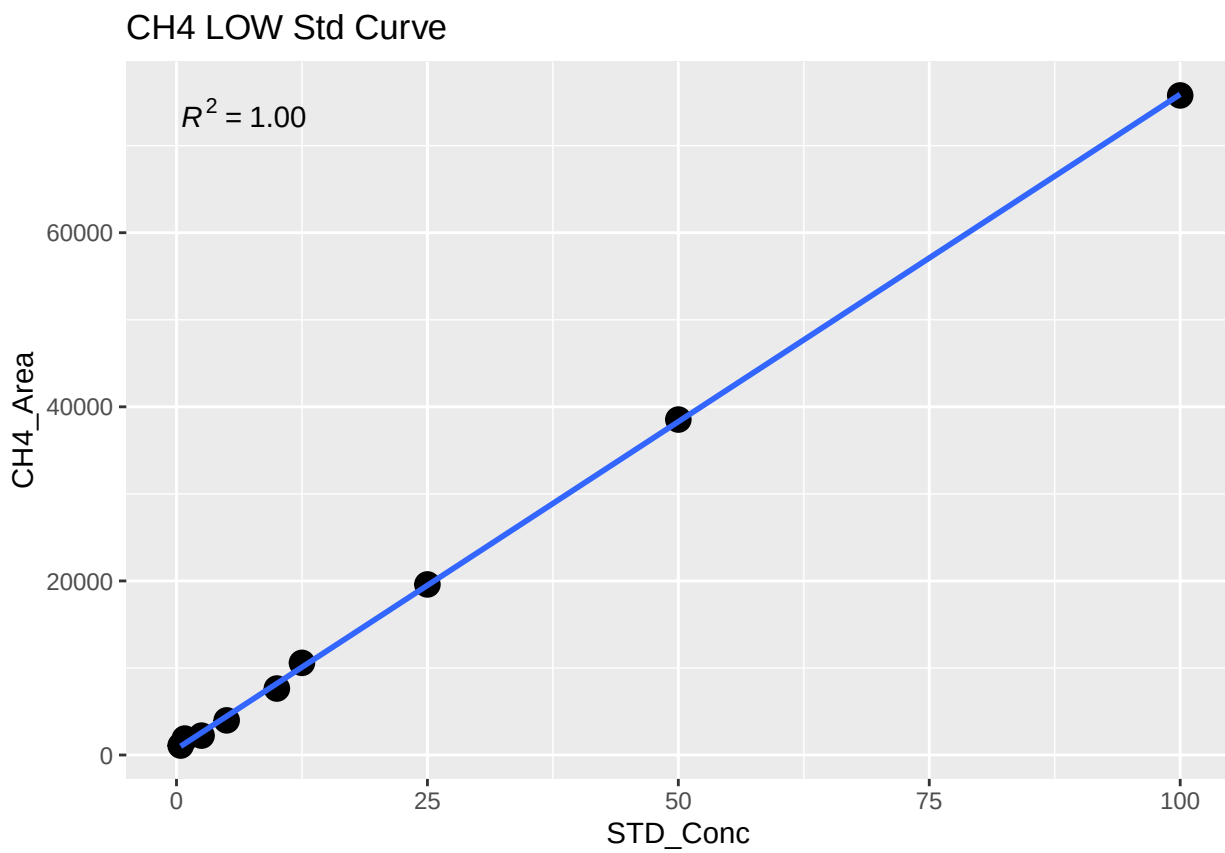
```
##     Machine    User Run_Date Sample_Type Sample1 Sample_Year Sample_Month
## 1 Varian GC Blakley 20260417     STD_CO2 STD_CO2          NA           NA
## 2 Varian GC Blakley 20260417     STD_CO2 STD_CO2          NA           NA
## 3 Varian GC Blakley 20260417     STD_CO2 STD_CO2          NA           NA
## 4 Varian GC Blakley 20260417     STD_CO2 STD_CO2          NA           NA
## 5 Varian GC Blakley 20260417     STD_CO2 STD_CO2          NA           NA
## 6 Varian GC Blakley 20260417     STD_CO2 STD_CO2          NA           NA
##   Sample_Day Sample_Time Time_Zone G_W Sample_Plot Sample_Grid Sample_Depth
## 1         NA          NA        NA  NA          NA          NA           NA
## 2         NA          NA        NA  NA          NA          NA           NA
## 3         NA          NA        NA  NA          NA          NA           NA
## 4         NA          NA        NA  NA          NA          NA           NA
## 5         NA          NA        NA  NA          NA          NA           NA
## 6         NA          NA        NA  NA          NA          NA           NA
##   Sample_ID Dilution_Factor    STD_Conc CO2_Area CH4_Area Field_Notes Lab_Notes
## 1      0ppm               1 0.000000000        0       NA          NA        NA
## 2    1mL_LT               1 0.008333333       NA       NA          NA        NA
## 3    2mL_LT               1 0.016666667       NA       NA          NA        NA
## 4    6mL_LT               1 0.050000000    63988       NA          NA        NA
## 5   12mL_LT               1 0.100000000   130481       NA          NA        NA
## 6   24mL_LT               1 0.200000000   261002       NA          NA        NA
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
## (`stat_smooth()`).
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
## (`stat_poly_eq()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```




```
##
## Call:
## lm(formula = stds_ch4_low$CH4_Area ~ stds_ch4_low$STD_Conc)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -572.07 -340.33   83.54  242.01  575.16
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)             688.089    180.870   3.804  0.00668 **
## stds_ch4_low$STD_Conc  752.098      4.685 160.530 9.61e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 436.7 on 7 degrees of freedom
##   (1 observation deleted due to missingness)
## Multiple R-squared:  0.9997, Adjusted R-squared:  0.9997
## F-statistic: 2.577e+04 on 1 and 7 DF,  p-value: 9.605e-14
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
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```

```
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## (`stat_poly_eq()`).

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## (`geom_point()`).
```
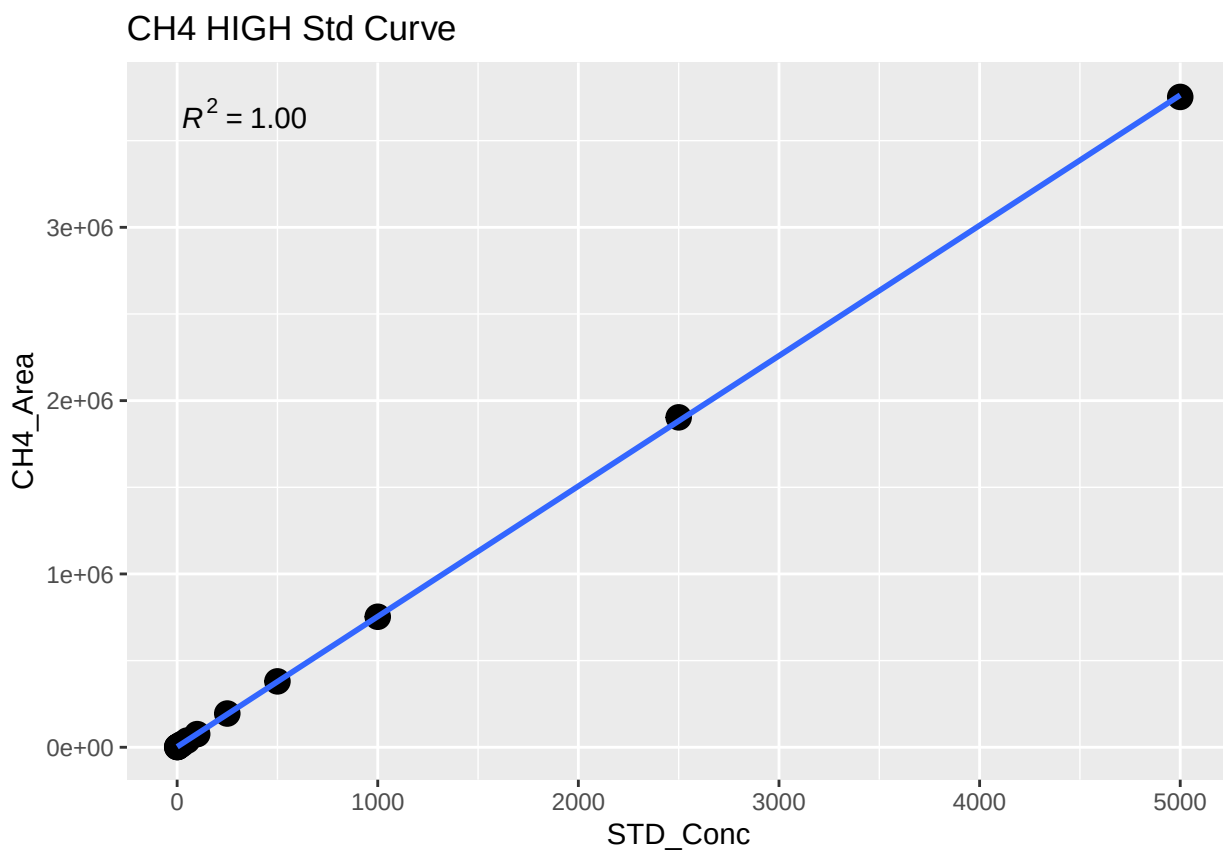
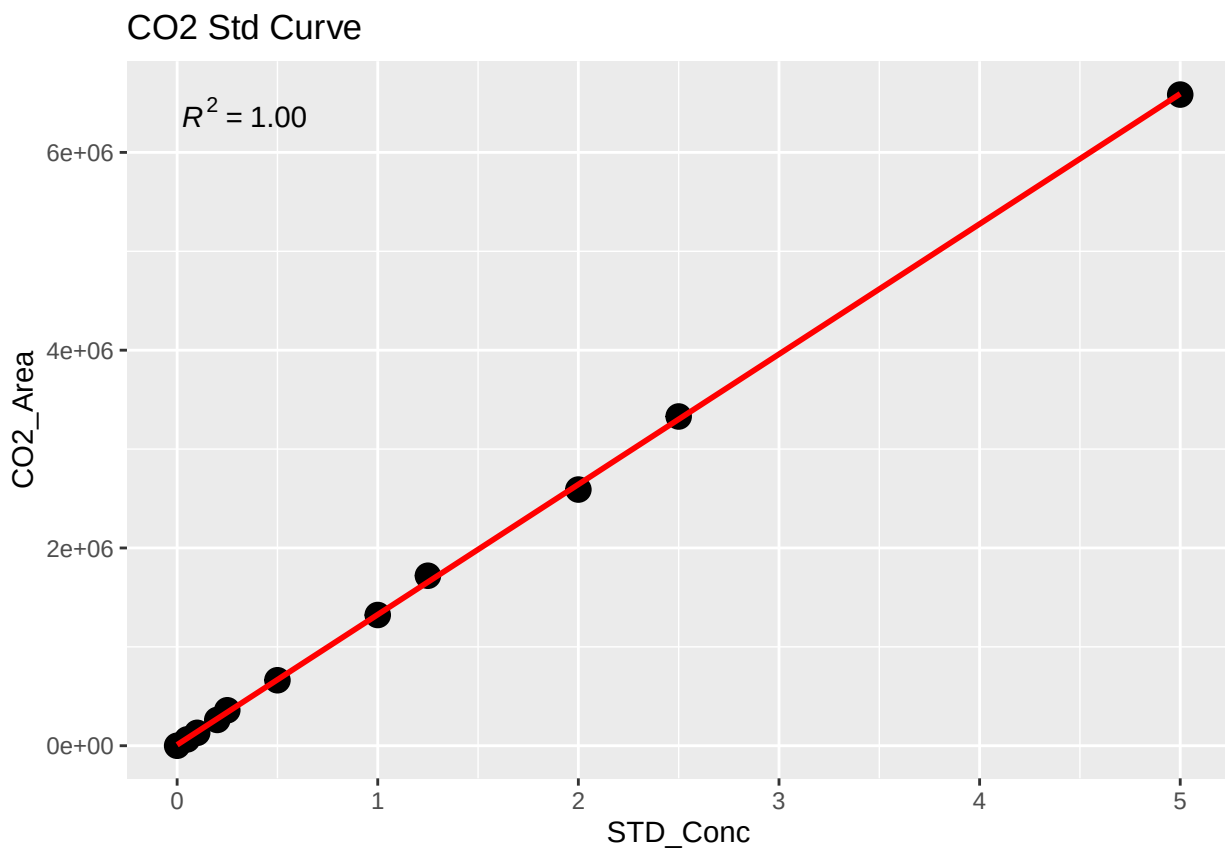



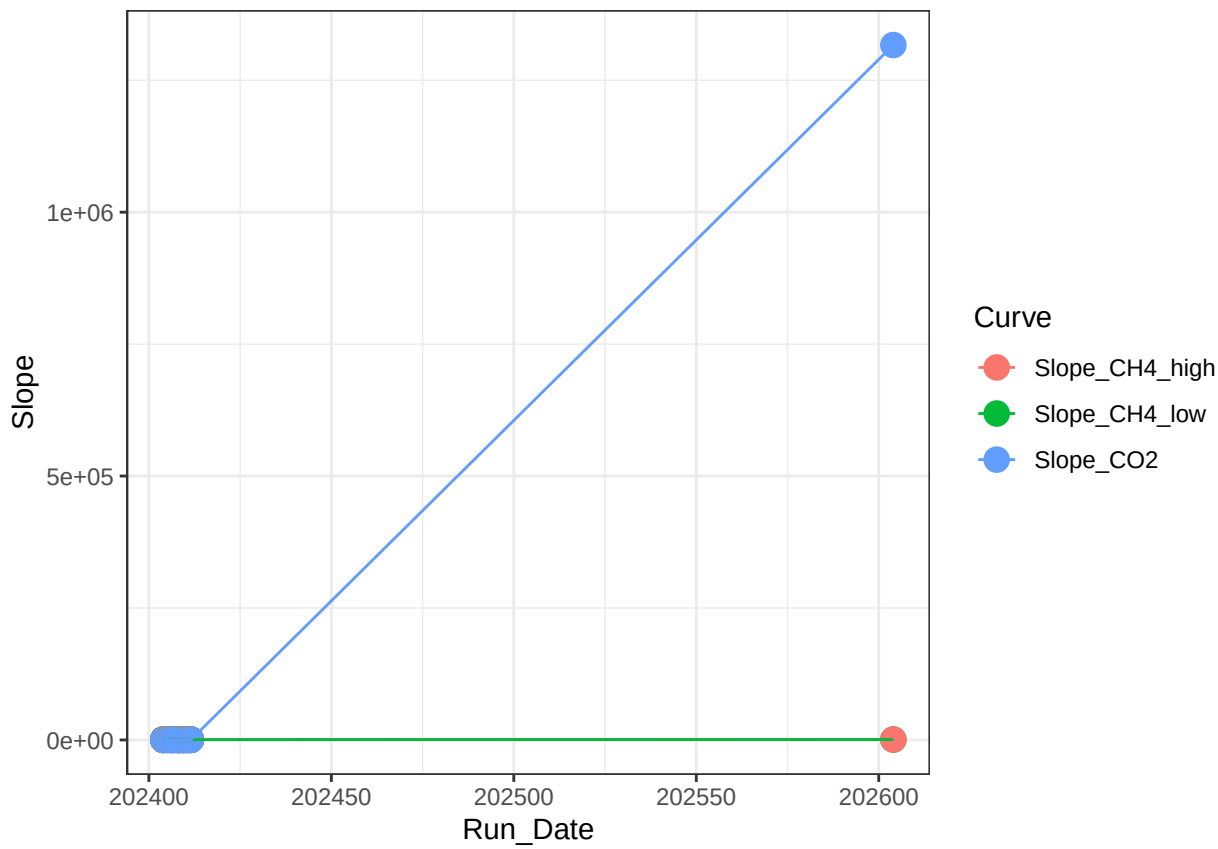

```
##
## Call:
## lm(formula = stds_ch4$CH4_Area ~ stds_ch4$STD_Conc)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -10484  -1944  -1510  -1101  20922
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)       2345.350   2101.477   1.116    0.286
## stds_ch4$STD_Conc  752.105      1.378 545.932   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7050 on 12 degrees of freedom
##   (1 observation deleted due to missingness)
## Multiple R-squared:      1,  Adjusted R-squared:      1
## F-statistic: 2.98e+05 on 1 and 12 DF,  p-value: < 2.2e-16
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 4 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 4 rows containing non-finite outside the scale range
## (`stat_poly_eq()`).

## Warning: Removed 4 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
##
## Call:
## lm(formula = stds_co2$CO2_Area ~ stds_co2$STD_Conc)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -52102 -10867  -8227   8066  63889
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)           9349      11955   0.782    0.454
## stds_co2$STD_Conc  1316705       6417 205.182   <2e-16 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 30890 on 9 degrees of freedom
##   (4 observations deleted due to missingness)
## Multiple R-squared:  0.9998, Adjusted R-squared:  0.9998
## F-statistic: 4.21e+04 on 1 and 9 DF,  p-value: < 2.2e-16

##   X          Curve        R2    Slope  Intercept Run_Date
## 1 1  Slope_CH4_low 0.9996407 633.0314  -320.6965   202404
## 2 2 Slope_CH4_high 0.9993443 601.5512 14817.1191   202404
## 3 3      Slope_CO2 0.9999907 168.3200 10075.2183   202404
## 4 4  Slope_CH4_low 0.9996407 633.0314  -320.6965   202404
## 5 5 Slope_CH4_high 0.9993443 601.5512 14817.1191   202404
## 6 6      Slope_CO2 0.9999907 168.3200 10075.2183   202404
```



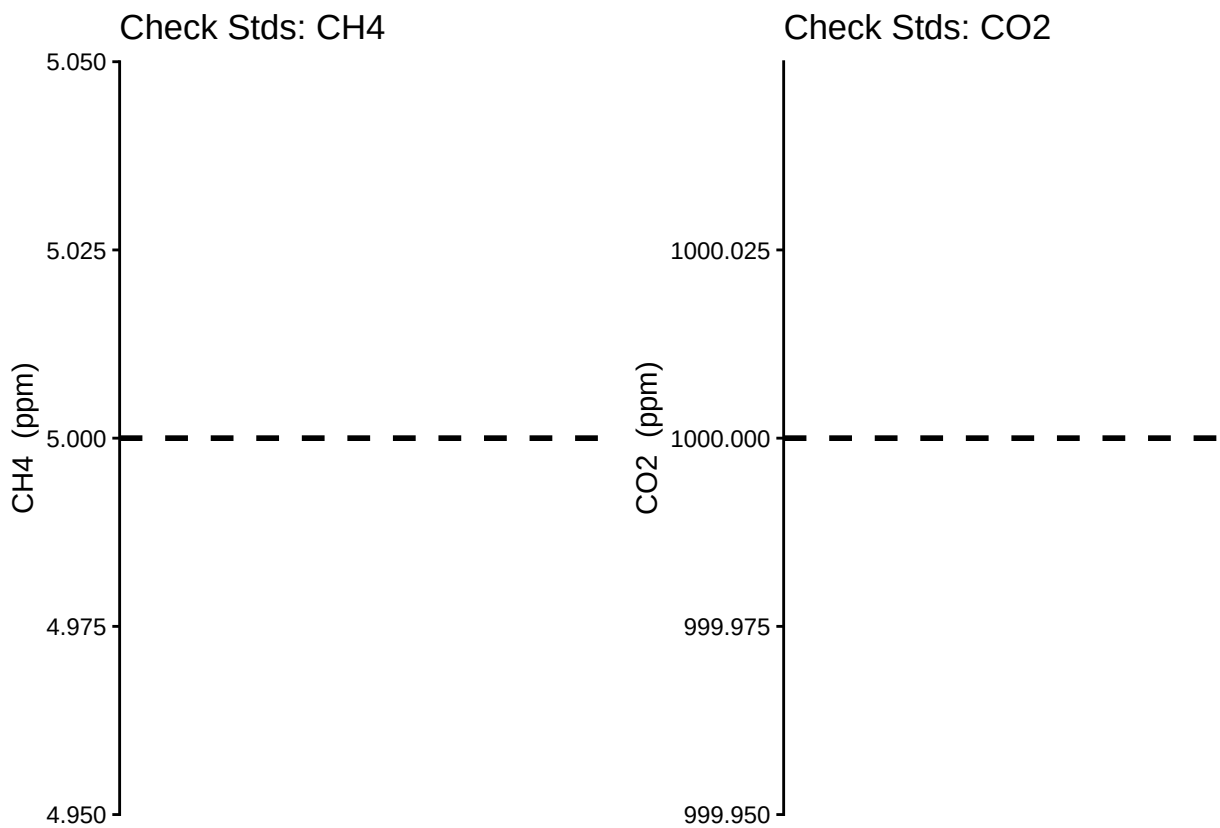
Now calculate the CH4 & CO2 concentrations in ppm

```
#head(raw)

#pull out methane standards
Samples <-  raw %>%
  filter(!str_detect(Sample_Type, "STD_CH4")) %>%
  filter(!str_detect(Sample_Type, "STD_CO2")) %>%
```

```r
  filter(!str_detect(Sample_Type, "Blank")) %>%
  filter(!str_detect(Sample_Type, "Chk_STD")) %>%
   filter(!str_detect(Sample_Type, "CHKSTD")) %>%
  filter(!str_detect(Sample_Type, "NA"))
#head(Samples)

#Now flag any areas that are above the 100ppm area for CH4
Samples$CH4_Curve <- ifelse(Samples$CH4_Area >71000, "High", "Low")
#head(Samples)

#Calculate CH4 concentrations in ppm
Samples$CH4_Conc_ppm <- ifelse(Samples$CH4_Area >71000, (Samples$CH4_Area-Slope_CH4_high$Intercept)/Slo

#Calculate CO2 concentrations in ppm
Samples$CO2_Conc_ppm <- ((Samples$CO2_Area-Slope_CO2$Intercept)/Slope_CO2$Slope)

#head(Samples)

##########make flags for any dilutions needed
#highest CH4 standard = 10000
#highest CO2 standard = 50000

Samples$CH4_Flag <- ifelse(Samples$CH4_Conc_ppm >10000, "Needs Dilution", "Within Range")
Samples$CO2_Flag <- ifelse(Samples$CO2_Conc_ppm >50000, "Needs Dilution", "Within Range")
#head(Samples)
```

Check the Check Standards

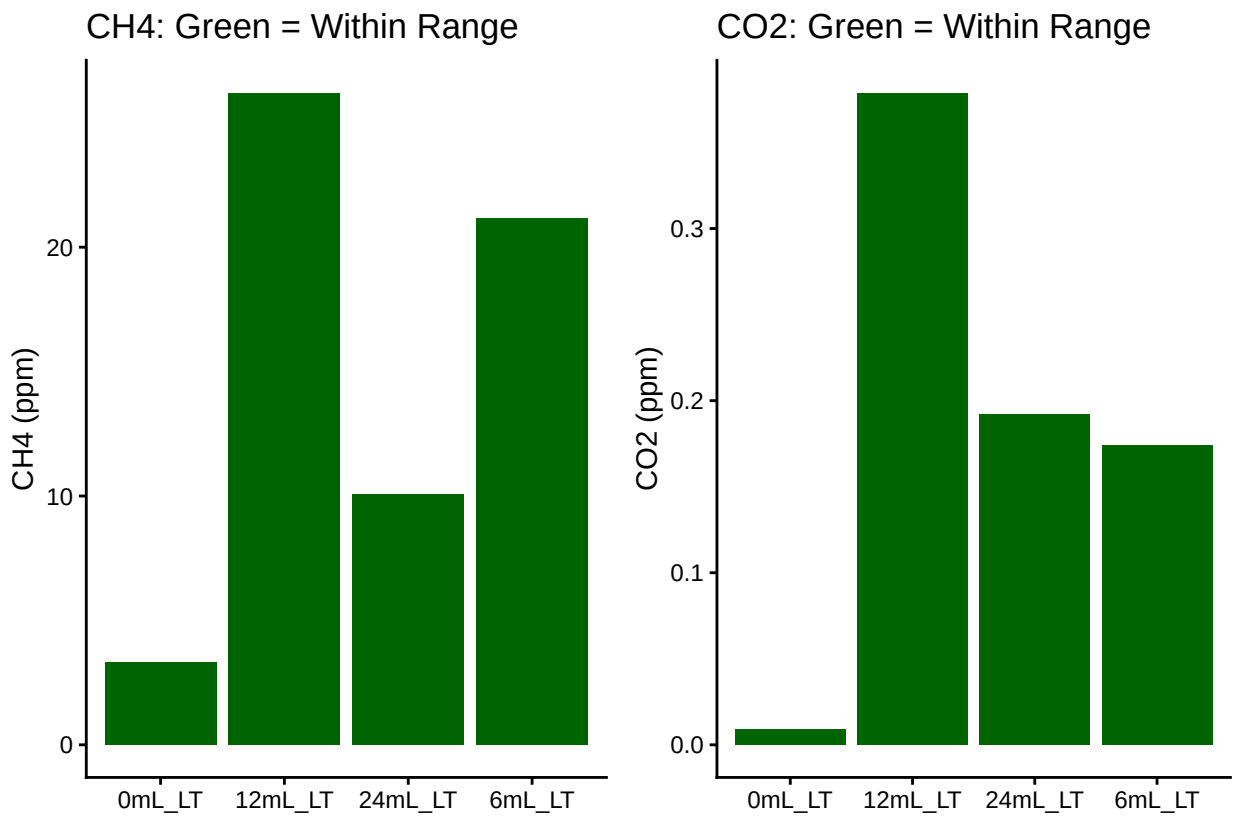
```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once per session.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```



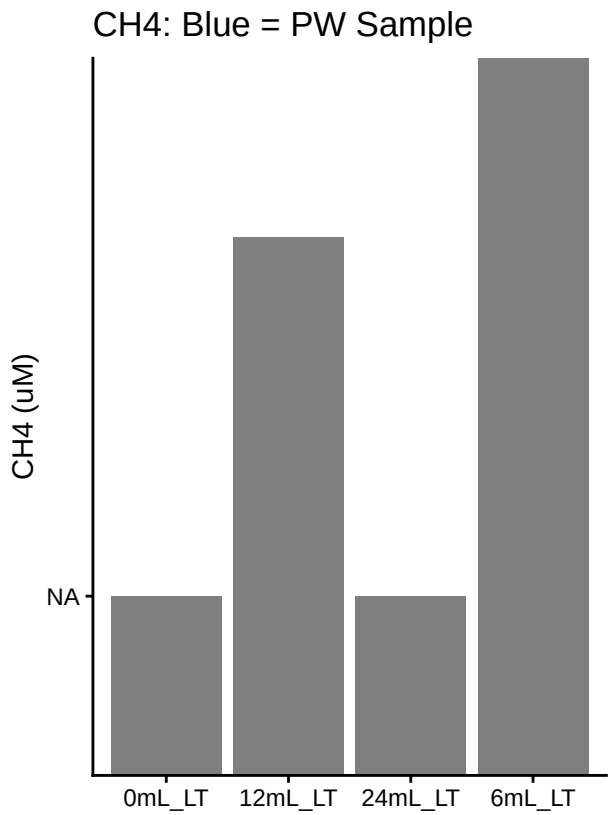
Dilution correct samples

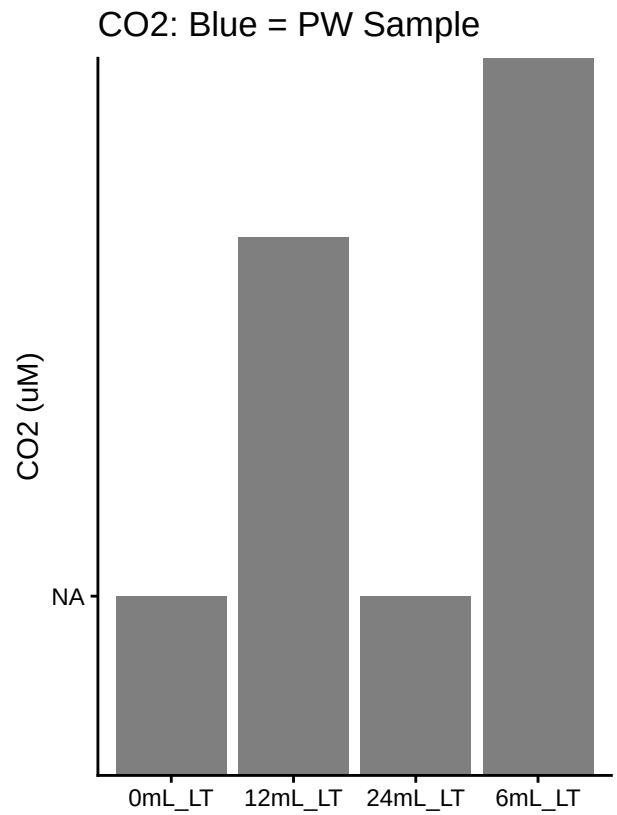
```
#multiply the concentration by the dilution factor
Samples$CH4_Conc_ppm_dilcorr <- (Samples$CH4_Conc_ppm * Samples$Dilution_Factor)

Samples$CO2_Conc_ppm_dilcorr <- (Samples$CO2_Conc_ppm * Samples$Dilution_Factor)

#check results
#head(Samples)

#quick first look at the samples
ch4_samples <-  ggplot(data = Samples, aes(x = Sample_ID, y = CH4_Conc_ppm, fill=CH4_Flag)) +
       geom_bar(stat = 'identity') +
        scale_fill_manual(values=c("darkgreen","red"))+
        #scale_fill_gradient2(low='red', mid='white', high='blue', space='Lab') +
        theme_classic() + labs(x= " ", y="CH4 (ppm)", title="CH4: Green = Within Range") +
        theme(legend.position="none")


co2_samples <-  ggplot(data = Samples, aes(x = Sample_ID, y = CO2_Conc_ppm, fill=CO2_Flag)) +
       geom_bar(stat = 'identity') +
        scale_fill_manual(values=c( "darkgreen", "red"))+
        #scale_fill_gradient2(low='red', mid='white', high='blue', space='Lab') +
        theme_classic() + labs(x= " ", y="CO2 (ppm)", title="CO2: Green = Within Range") +
```

```r
        theme(legend.position="none")

ggarrange(ch4_samples, co2_samples, nrow=1, ncol=2)
```



If samples are water calculate gas in water





Write out processed data & slopes - NOT THIS ONE

```
#check results
#head(Samples)

#pull out what we need
# Samples1 <- Samples[ ,c(1:3,6:16,20, 24:30)]
#head(Samples1)

#change the column names as needed
#colnames(samples1) <- c("Machine", "User", "Run_Date" , "Sample_Year" ,  "Sample_Month" , "Sample_Day"
 #                       "Sample_Time" , "Time_Zone" , "G_W" , "Sample_Plot","Sample_Grid","Sample_Dept
  #                      "Sample_ID" ,  "Dilution_Factor", "Notes" ,  "CH4_Flag", "CO2_Flag" ,
   #                     "CH4_Conc_ppm_dilcorr", "CO2_Conc_ppm_dilcorr")

write.csv(Samples, "Test_Run_Processed.csv")  #not writing out because curve is bad!!!!
```

##Combine sample data with expected concentrations

```
Exp_Conc <- read.csv("TMP_Low_Conc_Test.csv")
```

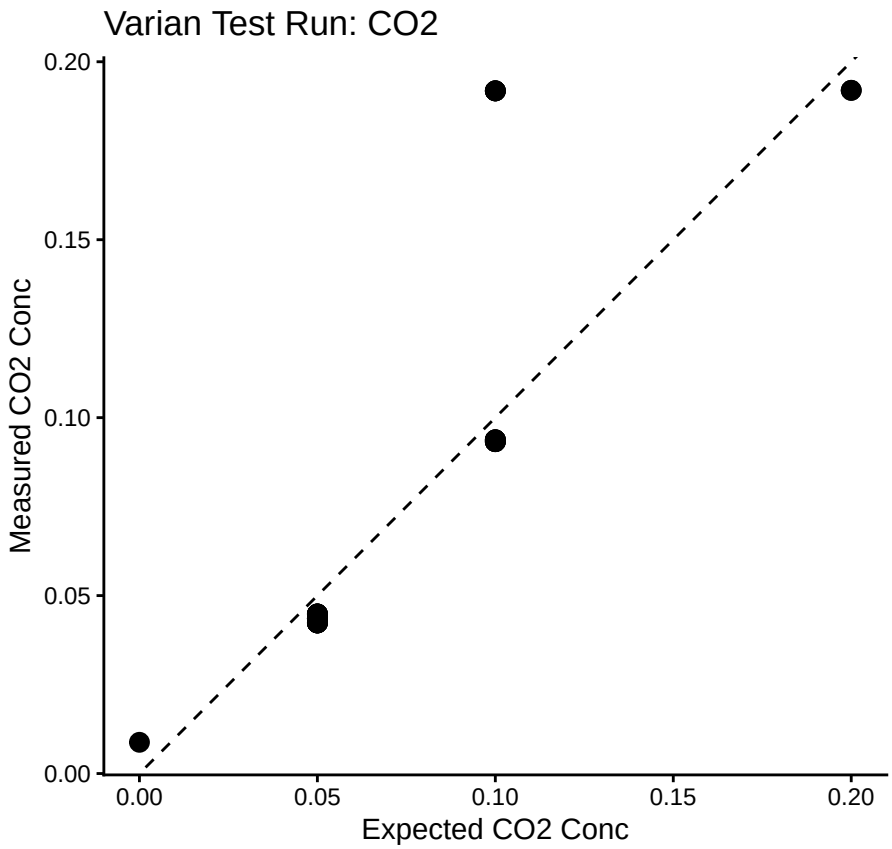
```r
Exp_Conc <- Exp_Conc %>%
  select(-c(CO2.Area, CH4.Area)) %>%
  rename(CO2_Conc_Exp = CO2.Conc....,
         CH4_Conc_Exp = CH4.Conc..ppm.)

Exp_Conc <- unique(Exp_Conc)


Samples2 <- left_join(Samples, Exp_Conc, by = c("Sample_ID" = "Standard"))
```

```
## Warning in left_join(Samples, Exp_Conc, by = c(Sample_ID = "Standard")): Detected an unexpected many-
## i Row 1 of ‘x‘ matches multiple rows in ‘y‘.
## i Row 4 of ‘y‘ matches multiple rows in ‘x‘.
## i If a many-to-many relationship is expected, set ‘relationship =
##   "many-to-many"‘ to silence this warning.
```

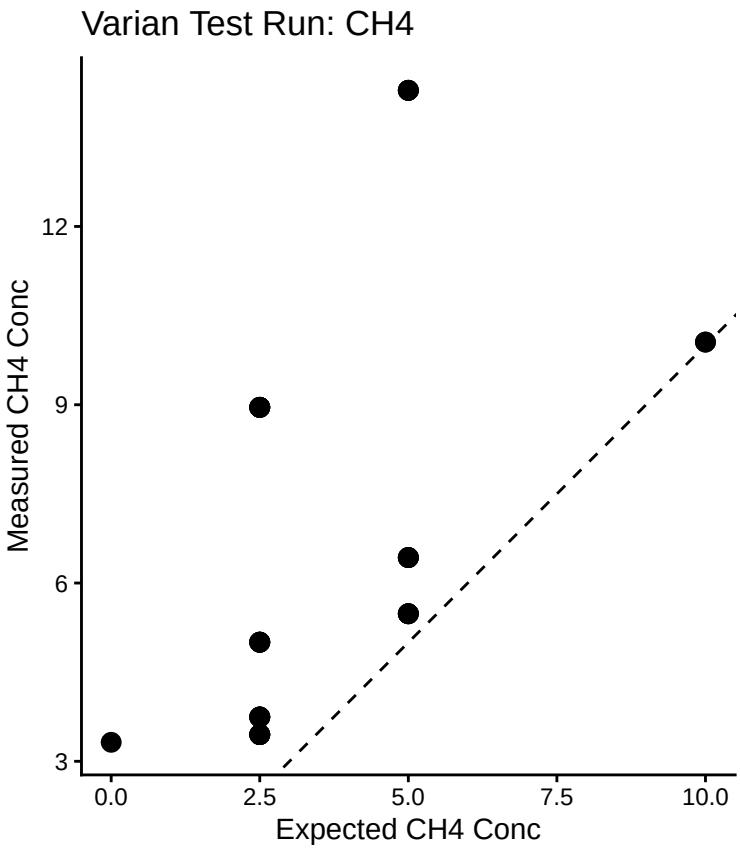
```r
Samples2 <- Samples2 %>%
  select(Sample_ID, CO2_Conc_ppm, CH4_Conc_ppm, CO2_Conc_Exp, CH4_Conc_Exp)
```

##Plot Expected vs Actual Conc

```r
##Plot CO2 Data
CO2_plots <- ggplot(Samples2, aes(x = CO2_Conc_Exp, y = CO2_Conc_ppm)) +
  geom_point(size = 3) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed") +
  # coord_cartesian(xlim = combined_limits, ylim = combined_limits) +
  coord_fixed(ratio = 1) +
  labs(
    y = "Measured CO2 Conc",
    x = "Expected CO2 Conc",
    title = "Varian Test Run: CO2"
  ) +
  theme_classic()

CO2_plots
```




```
##Plot CH4 Data
CH4_plots <- ggplot(Samples2, aes(x = CH4_Conc_Exp, y = CH4_Conc_ppm)) +
  geom_point(size = 3) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed") +
  # coord_cartesian(xlim = combined_limits, ylim = combined_limits) +
  coord_fixed(ratio = 1) +
  labs(
    y = "Measured CH4 Conc",
    x = "Expected CH4 Conc",
    title = "Varian Test Run: CH4"
  ) +
  theme_classic()

CH4_plots
```



##Put Samples in Order and Plot in Order